

JINHAO LIANG

RESEARCH INTERESTS

I am particularly interested in integrating generative modeling with optimization to address complex engineering and scientific challenges. I am currently developing algorithms to ensure that the outputs of diffusion and flow-based generative models satisfy constraints with provable guarantees, with the goal of enabling scalable and adaptive solutions in robotics, biology, and physics.

EDUCATION

University of Virginia (UVA) Ph.D. in Computer Science Advisor: Prof. Ferdinando Fioretto	Sep. 2024 - Present GPA: 3.77/4.00
The Chinese University of Hong Kong, Shenzhen (CUHKSZ) M.Phil in Computer and Information Engineering Advisor: Prof. Chenye Wu	Sep. 2022 - Jul. 2024 GPA: 3.90/4.00
Xidian University (XDU) B.E. in Software Engineering	Sep. 2018 - Jun. 2022 GPA: 3.70/4.00

EXPERIENCE

University of Virginia Ph.D. Student / Graduate Research Assistant Mentor: Ferdinando Fioretto	Sep. 2024 - Present
Argonne National Laboratory Givens Associate Mentor: Sandeep Madireddy	Jun. 2026 - Aug. 2026
Argonne National Laboratory Research Aide Mentor: Sandeep Madireddy	Jun. 2025 - Aug. 2025

PUBLICATIONS

Machine Learning Conference Publications

- Jinhao Liang**, Sven Koenig and Ferdinando Fioretto, “Simulation-Informed Diffusion for Decentralized Multi-robot Motion Planning.” accepted by *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- Jinhao Liang**, Yixuan Sun, Anirban Samaddar, Sandeep Madireddy and Ferdinando Fioretto, “Chance-constrained Flow Matching for High-Fidelity Constraint-aware Generation.” accepted by *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- Prince Zizhuang Wang, Shuyi Chen, **Jinhao Liang**, Ferdinando Fioretto, and Shixiang Zhu, “Gen-dfl: Decision-focused generative learning for robust decision making.” accepted by *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
🏆 IISE DAIS Best Student Paper Finalist.
- Jinhao Liang**, Sven Koenig and Ferdinando Fioretto, “Discrete-Guided Diffusion for Scalable and Safe Multi-Robot Motion Planning.” accepted by *The 40th Annual AAAI Conference on Artificial Intelligence (AAAI)*, 2025.

5. Jacob K. Christopher, Michael Cardei, **Jinhao Liang**, and Ferdinando Fioretto, “Neuro-Symbolic Generative Diffusion Models for Physically Grounded, Robust, and Safe Generation.” accepted by *The 2nd International Conference on Neuro-symbolic Systems (NeuS)*, 2025.
6. **Jinhao Liang**, Jacob K. Christopher, Sven Koenig and Ferdinando Fioretto, “Simultaneous Multi-Robot Motion Planning with Projected Diffusion Models.” accepted by *The 42nd International Conference on Machine Learning (ICML)*, 2025.
7. **Jinhao Liang**, Jacob K. Christopher, Sven Koenig and Ferdinando Fioretto, “Multi-Agent Path Finding in Continuous Spaces with Projected Diffusion Models.” accepted by *The 6th International Workshop on Multi-Agent Path Finding, at AAAI*, 2025.

Journal Articles

1. **Jinhao Liang**, Wenqian Jiang, Chenbei Lu and Chenye Wu, “Joint Chance-constrained Unit Commitment: Statistically Feasible Robust Optimization with Learning-to-Optimize Acceleration.” accepted by *IEEE Transactions on Power Systems*, 2024. (JCR Q1)
2. Chenbei Lu, **Jinhao Liang**, Nan Gu, Haoxiang Wang and Chenye Wu, “Manipulation-Proof Virtual Bidding Mechanism Design.” accepted by *IEEE Transactions on Energy Markets, Policy and Regulation*, 2023.
3. Chenbei Lu, **Jinhao Liang**, Wenqian Jiang, Jiaye Teng and Chenye Wu, “High-Resolution Probabilistic Load Forecasting: A Learning Ensemble Approach.” accepted by *Journal of the Franklin Institute*, 2023. (JCR Q1)

Power & Control Conference Publications

1. Chenbei Lu, **Jinhao Liang**, Hongyu Yi and Chenye Wu, “Cost-effective Closed-loop Bilevel Robust Optimization for Joint Chance-constrained Economic Dispatch.” accepted by *The 16th ACM International Conference on Future and Sustainable Energy Systems (e-Energy)*, 2025.
2. **Jinhao Liang**, Ferdinando Fioretto and Chenye Wu, “Robust Bidding Strategies in Local Energy Markets.” accepted by *2025 IEEE Power & Energy Society General Meeting (PESGM)*, 2025.
3. **Jinhao Liang**, Chenbei Lu, Wenqian Jiang and Chenye Wu, “Few-shot Residential Load Forecasting Boosted by Learning to Ensemble.” accepted by *the 7th IEEE Conference on Energy Internet and Energy System Integration (EI²)*, 2023.
4. Wenqian Jiang, **Jinhao Liang**, Chenbei Lu and Chenye Wu, “Robust Online EV Charging Scheduling with Statistical Feasibility.” accepted by *the 62nd IEEE Conference on Decision and Control (CDC)*, 2023.
5. **Jinhao Liang**, Wenqian Jiang and Chenye Wu, “Effective Carbon Tax Learning via Cap and Trade.” accepted by *IEEE 5th International Electrical and Energy Conference (CIEEC)*, 2022.

SELECTED RESEARCH EXPERIENCE

Chance-constrained Flow Matching for High-Fidelity Constraint-aware Generation

Supervisor: Prof. Ferdinando Fioretto and Sandeep Madireddy

Feb. 2025 - Jul. 2025

- Integrated stochastic optimization into flow-matching generative models, enabling **concise and effective enforcement of hard constraints while preserving the high-fidelity sample generation.**

- Established a theoretical framework for enforcing hard constraints in the sampling process of generative models, demonstrating the **feasibility guarantees and sample quality preservation theoretically**.
- Achieved **2.3× higher feasibility** and over **20× higher efficiency** than prior constrained generative models.

Discrete-guided Diffusion for Scalable and Safe Multi-Robot Motion Planning

Supervisor: Prof. Ferdinando Fioretto

Feb. 2025 - Jul. 2025

- Decomposed the original MRMP problem into a sequence of local subproblems with convex configuration spaces.
- Guided diffusion models with spatiotemporal structure derived from the MAPF solution to generate high-quality trajectories.
- Designed a lightweight projection mechanism that enforces feasibility and activates only when constraint violations are detected.

Multi-Robot Motion Planning with Projected Diffusion Models

Supervisor: Prof. Ferdinando Fioretto

Sep. 2024 - Jan. 2025

- Introduced a novel formulation of Multi-Robot Motion Planning (MRMP) in continuous spaces using diffusion models, enabling the simultaneous generation of trajectories for all agents in a single framework.
- Adapted projected diffusion models (PDM) for MRMP by embedding constraints directly into the diffusion process, ensuring that the generated solutions are feasible and collision-free.
- Developed an augmented Lagrangian approach to accelerate the projection process, making the method scalable and practical for real-world applications.

Joint Chance-constrained Unit Commitment with Statistical Feasibility

Supervisor: Prof. Chenye Wu

Oct. 2022 - Jun. 2023

- Extended the notion of statistical feasibility into unit commitment, a mixed-integer problem, and formulated the statistically feasible unit commitment.
- Developed sample-based uncertainty set construction algorithms, yielding less conservative solutions.
- Accelerated the solving process and designed the optimization kernel to boost its computational efficiency further.

SELECTED TALKS

“Constrained Flow Matching for Scientific Applications”

Mar. 2026

Minisymposium on *Application of Flow-based Generative Models in Science*, SIAM UQ26.

ACADEMIC SERVICES

Reviewer

- **Journals:** IEEE Transactions on Robotics, IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE Transactions on Energy Markets, Policy, and Regulation, IEEE Transactions on Emerging Topics in Computational Intelligence, Applied Energy, and Transactions on Machine Learning Research.
- **Conferences:** AAAI, NeurIPS, ICML, ICLR, AAMAS, and CoRL.

SELECTED HONORS

NeurIPS 2026 Top Reviewer	Oct 2026
ICML 2026 Silver Reviewer Recognition	May 2026
FPI AITHYRA Travel Grant	Nov 2025
DARPA Disruptive Idea Award	May 2025
WoMAPF student travel grants	Dec 2024
AAAI Student Scholarship	Dec 2024
Presidential Award for Outstanding Graduate Students (top 10 in CUHKSZ)	Nov 2024
Provost Fellowship (top graduate students in UVA)	Sep 2024